

Week 03

1) Population and Sample

- * Population means the entire group of individuals or elements you want to study which share a common behaviour

$$\text{Population Size} = N$$

- * Sample means the subset of the population you use to draw conclusions about the population as a whole.

$$\text{Sample Size} = n$$

2) Sample mean

- * Sample mean is not equal to the actual population mean.
- * But bigger the sample size it is close to the actual population mean.

3) Sample proportion

$$\text{Population proportion} = \frac{\text{number of items with a given characteristic } (x)}{\text{population } (N)}$$

(P)

$$(\hat{P}) \text{ Sample proportion} = \text{estimate of the population proportion}$$

4) Sample Variance

$$\text{Var}(x) = \frac{1}{N} \sum (x - \mu)^2 \rightarrow \text{var}(\hat{X}) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

This equation is biased
It underestimates the
population variance

$$S^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

$\bar{x}$ = Sample mean

* $n-1$ fixes bias when all you have is a sample.

* As n gets big, the difference matters less.

* If it matters, you may have too little data

Population Variance
Formula

$$\text{Var}(X) = \sigma^2 =$$

$$\frac{1}{N} \sum (x - \mu)^2$$

Sample Variance
Formula

$$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

(most common)

$$\sigma^2 \approx \text{Var}(\hat{X}) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

5) Law of large numbers

* As the sample size increases, the average of the sample will tend to get closer to the average of the entire population.

n : number of samples

X_i : is the i th random sample from the population.

Each X_i are independent and identically distributed (i.i.d.).

$$\text{as } n \rightarrow \infty \quad \frac{1}{n} \sum_{i=1}^n X_i \rightarrow E[X] = \mu$$

$$\text{as } n \rightarrow \infty \quad \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \rightarrow E[X] = \mu$$

⇓

Under Certain conditions

- * Sample is randomly drawn.
- * Sample size must be sufficiently large.
- * Independent observations.

6) Central Limit Theorem (Discrete)

* If you take the sum (or average) of a large number of independent and identically distributed (i.i.d) random variables, then their distribution will approximate a Normal distribution, regardless of the original distribution as long as the variance is finite.

$X_1, X_2, \dots, X_n$ (i.i.d. discrete random variables)

$$\mu = E[X] \quad \text{Variance: } (\sigma^2) = \text{Var}(X)$$

* Sample mean :

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

* CLT statement

As $n \rightarrow \infty$:

$$\frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}} \xrightarrow{d} N(0, 1)$$

* This means the standardized sample mean tends to a standard normal distribution.

* As n become sufficiently large we will get a normal distribution with parameters

$$\begin{aligned} \mu &= np \\ \sigma^2 &= np(1-p) \end{aligned}$$

1) Central Limit Theorem (Continuous)

2) When you average a large enough number of variables, the distribution will approximately follow a normal distribution.

$$\text{As } n \rightarrow \infty \quad \frac{\frac{1}{n} \sum_{i=1}^n X_i - E[X]}{\sigma_X} \sqrt{n} \sim N(0, 1^2)$$

$$\text{As } n \rightarrow \infty \quad \frac{\sum_{i=1}^n X_i - n E[X]}{\sqrt{n} \sigma_X} \sim N(0, 1^2)$$

$\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_k$

means of
sample datasets

(μ, σ^2)

No matter the
distribution



Converts
to
Normal
distribution



The more samples $\rightarrow k \rightarrow \infty$
you extract

The bigger the $\rightarrow n \rightarrow \infty$
samples

$$N \sim \left(\mu, \frac{\sigma^2}{n} \right)$$

8) Maximum Likelihood Estimation

* MLE is a method to estimate the parameters of a statistical model.

* It finds the parameter values that make the observed data most likely

model 1 $\frac{P(\text{DATA}/m_1)}$
 model 2 $\frac{P(\text{DATA}/m_2)}$
 model 3 $\frac{P(\text{DATA}/m_3)}$

DATA

Find model that likely produced the data
 (Maximize $P(\text{DATA}/\text{model})$)

9) MLE (Bernouli Example)

n coins
 k heads

$x_1 \quad x_2 \quad x_3 \quad \dots \quad x_{n-1} \quad x_n$

$$X = (x_1, \dots, x_n)$$

$X_i \stackrel{\text{i.i.d.}}{\sim} \text{Bernouli}(p)$

Likelihood

$$L(p; x) = P_p(X=x) = \prod_{i=1}^n P_{x_i}(x_i) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}$$

if $x_i = 1, p^{x_i} (1-p)^{1-x_i} = p$
 if $x_i = 0, p^{x_i} (1-p)^{1-x_i} = (1-p)$

$$\sum_{i=1}^n x_i = \# \text{ heads}$$

$$n - \sum_{i=1}^n x_i = \# \text{ tails}$$

$$L = p \left(\sum_{i=1}^n x_i \right) (1-p)^{n - \sum_{i=1}^n x_i}$$

Log-likelihood

$$\begin{aligned} L(p; x) &= \log \left(p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i} \right) \\ &= \left(\sum_{i=1}^n x_i \right) \log(p) + \left(n - \sum_{i=1}^n x_i \right) \log(1-p) \end{aligned}$$

Find the maximum

$$\frac{d}{dp} L(p; x) = \frac{\sum_{i=1}^n x_i}{p} + \frac{n - \sum_{i=1}^n x_i}{1-p} (-1) = 0$$

$$\hat{p} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}$$

k heads $\rightarrow$ optimal probability to obtain those k heads would be precisely k over n .

$$L(p; 8H) = p^8 (1-p)^2 \Rightarrow \log(p; 8H) = \log(p^8 (1-p)^2) = 8 \log(p) + 2 \log(1-p)$$

$$\frac{d}{dp} (8 \log(p) + 2 \log(1-p)) = \frac{8}{p} + \frac{2}{1-p} (-1)$$

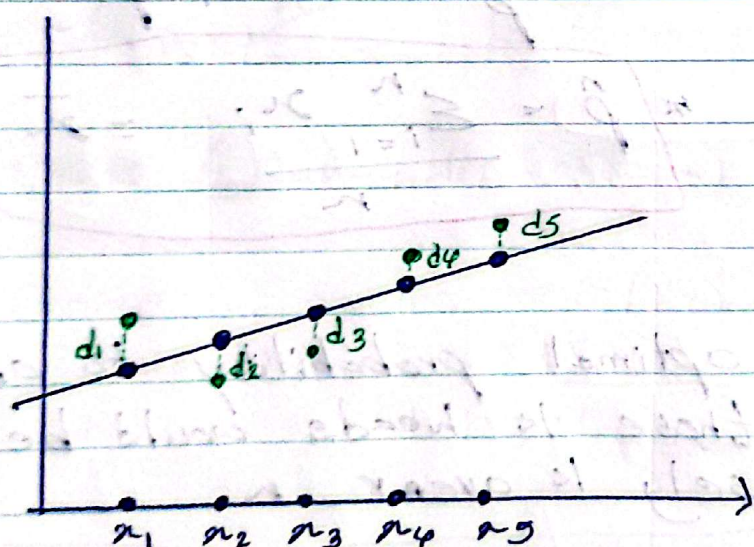
$= 0$

$$\hat{p} = \frac{1.8}{10} //$$

10) MLE Gaussian problem

- * The best distribution is the one where the mean of the distribution is the mean of the sample ($\mu = \mu'$)
- * The best distribution is the one where the variance of the distribution is the variance of the sample ($\sigma^2 = \sigma'^2$)

11) MLE linear Regression



Likelihood $(N(0,1))$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_2^2}$$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_4^2}$$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_5^2}$$

Maximize $\Rightarrow e^{-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2)}$

Minimize $\Rightarrow d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$

Least square Error = Linear Regression

12) Regularization

* Regularization is a technique to prevent overfitting by adding a penalty to the model's loss function.

$$\text{Model} \Rightarrow y = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

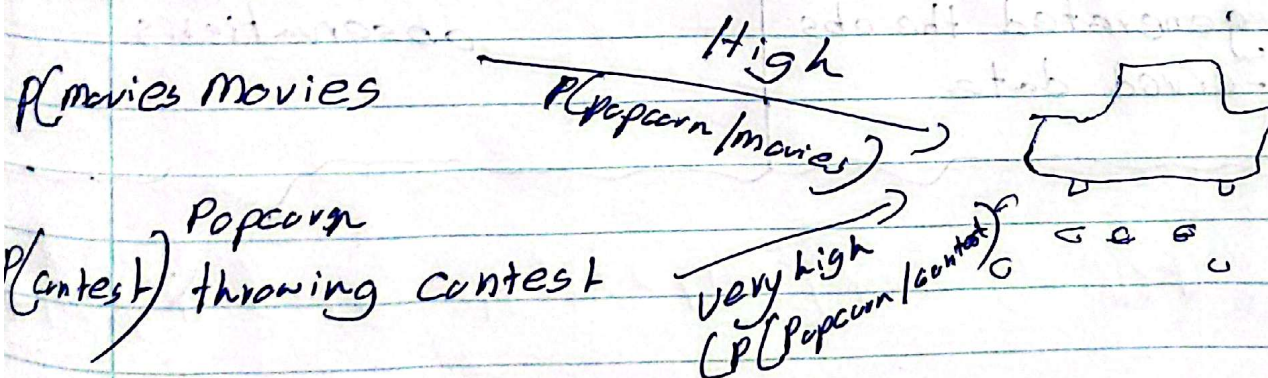
$$\text{Log loss} \Rightarrow ll$$

$$\text{L2 Regularization Error} \Rightarrow a_n^2 + a_{n-1}^2 + \dots + a_1^2$$

$$\text{Regularization parameter} \Rightarrow \lambda$$

$$\text{Regularized Error} = ll + \lambda (a_n^2 + a_{n-1}^2 + \dots + a_1^2)$$

13) Bayesics



$$P(\text{Popcorn} | \text{movies}) \cdot P(\text{movies}) > P(\text{popcorn} | \text{contest}) \cdot P(\text{contest})$$

$$P(A|B) \cdot P(B) = P(A \cap B)$$

14) Bayesian vs Frequentist

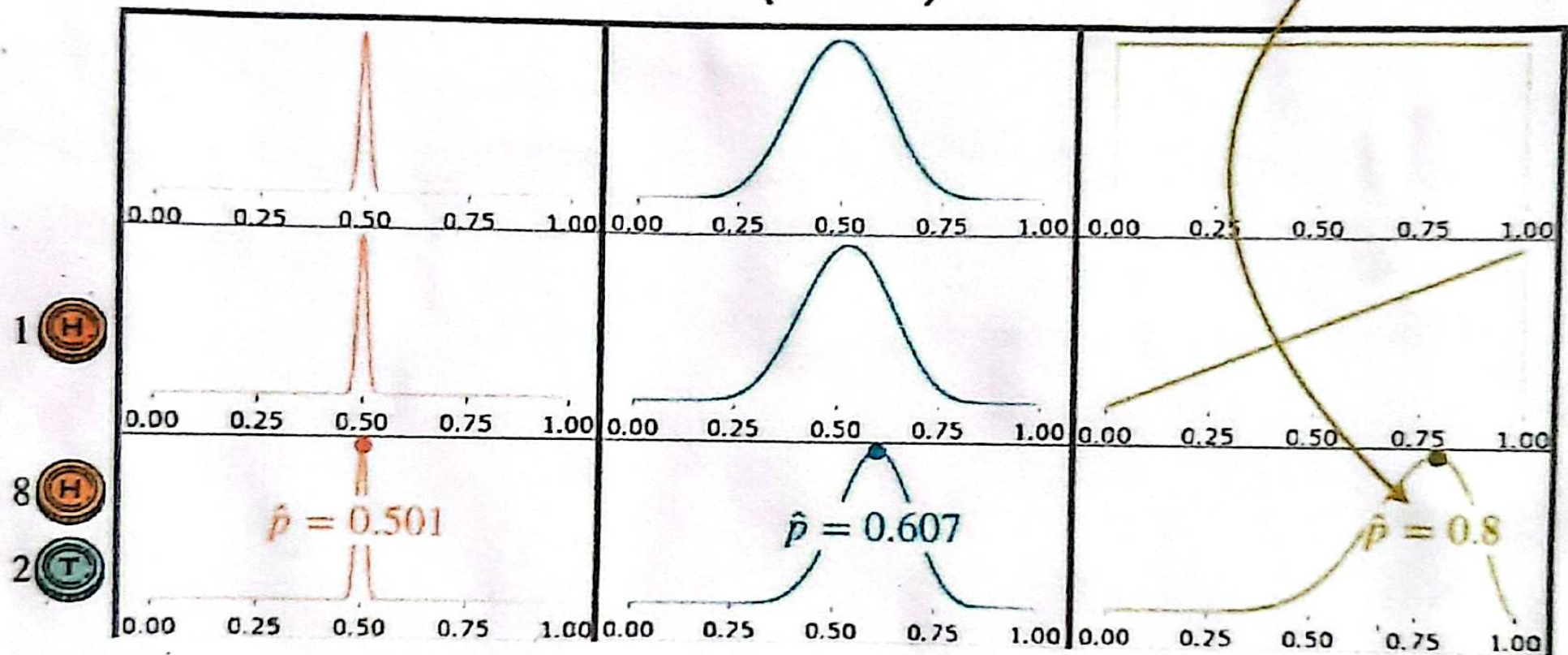
* Frequentist $\Rightarrow$ Either it works or it doesn't, we will test it many times and see how often it succeeds.

* Bayesian $\Rightarrow$ I start with some belief (say, 50% chance it works) and update the belief as I collect results

Frequentists	Bayesians
* Probabilities represent long term frequency of events.	* Probabilities represent the degree of belief (or certainty)
* Concept of likelihood	* Concept of prior
* Goal: Find the model that most likely generated the observed data	* Goal: update prior belief based on observations

Maximum a Posteriori (MAP)

Same result as frequentist!



• $P(B|A)$ = Likelihood of evidence B appearing, given A happened

• $P(A)$ = "Prior" Belief that A will happen before considering evidence B

• $P(B)$ = Probability of evidence B in any circumstances

$A \Rightarrow$ an event you are trying to predict
(you are offered a job)

$B \Rightarrow$ another event or evidence that helps refine your prediction (you were asked to a follow up phone call)

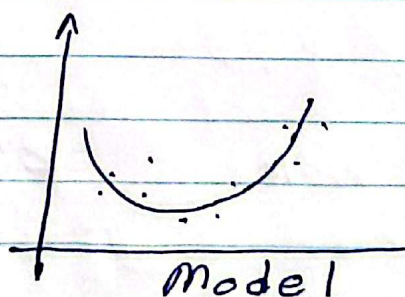
	Y is discrete	Y is Continuous
X is discrete	$P_{Y X=x}(y) = \frac{P_{X Y=y}(x) P_Y(y)}{P_X(x)}$	$f_{Y X=x}(y) = \frac{P_{X Y=y}(x) f_Y(y)}{P_X(x)}$
X is Continuous	$P_{Y X=x}(y) = \frac{f_{X Y=y}(x) P_Y(y)}{f_X(x)}$	$f_{Y X=x}(y) = \frac{f_{X Y=y}(x) f_Y(y)}{f_X(x)}$

Summary

- * Bayesians update prior beliefs
- * MAP with uninformative priors is just MLE
- * With Enough data, MLE and MAP estimates usually converge
- * Good for instances where you have limited data or strong prior beliefs
- * Wrong priors, wrong decisions

1) Relationship between MAP, MLE and Regularization

Maximum Likelihood
with Bayes



Polynomial regression
with regularization

$$P(\text{Data}|\text{Model})$$

$$P(\text{Model})$$

Take
Logarithms

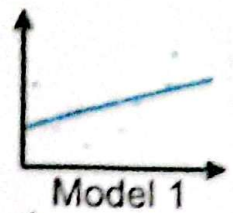
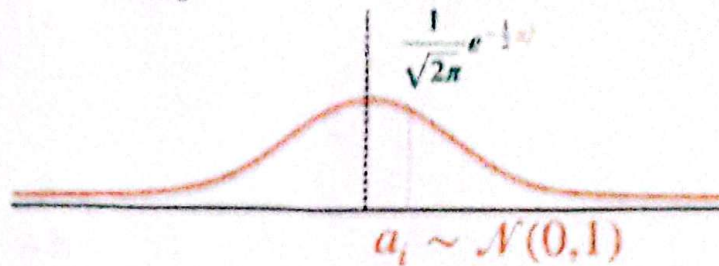
Square loss
+

Regularization Term

What Is the Probability of a Model?

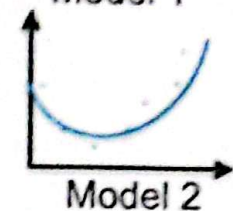
$$P(\text{Model 1}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2}$$

$$a_1 x + b$$



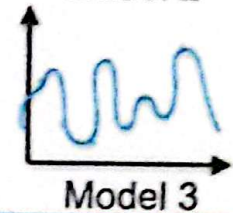
$$P(\text{Model 2})$$

$$a_1 x^2 + a_2 x + b = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$



$$P(\text{Model 3})$$

$$a_1 x^{10} + a_2 x^9 + \dots + a_{10} x + b = \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_i^2}$$



Bayes and Regularization

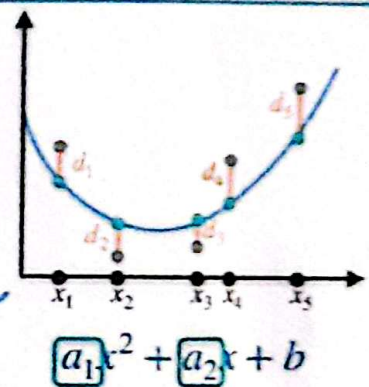
$$P(\text{Data}|\text{Model}) \quad P(\text{Model})$$

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_5^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2}$$

Maximize

$$\log \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_3^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_4^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_5^2} \right) + \log \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_1^2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}a_2^2} \right)$$

$$-\frac{1}{2}(d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2) + -\frac{1}{2}(a_1^2 + a_2^2)$$



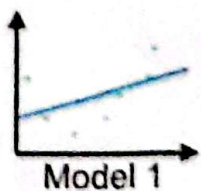
Minimize $d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + a_1^2 + a_2^2$

Regularization

$P(\text{Model 1})$ **New Loss**

Minimize $a_1^2 + d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$

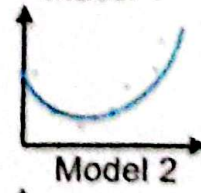
$P(\text{Data}|\text{Model 1})$



$P(\text{Model 2})$

Minimize $a_1^2 + a_2^2 + d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$

$P(\text{Data}|\text{Model 2})$



$P(\text{Model 3})$

Minimize $a_1^2 + \dots + a_{10}^2 + d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$

$P(\text{Data}|\text{Model 3})$

